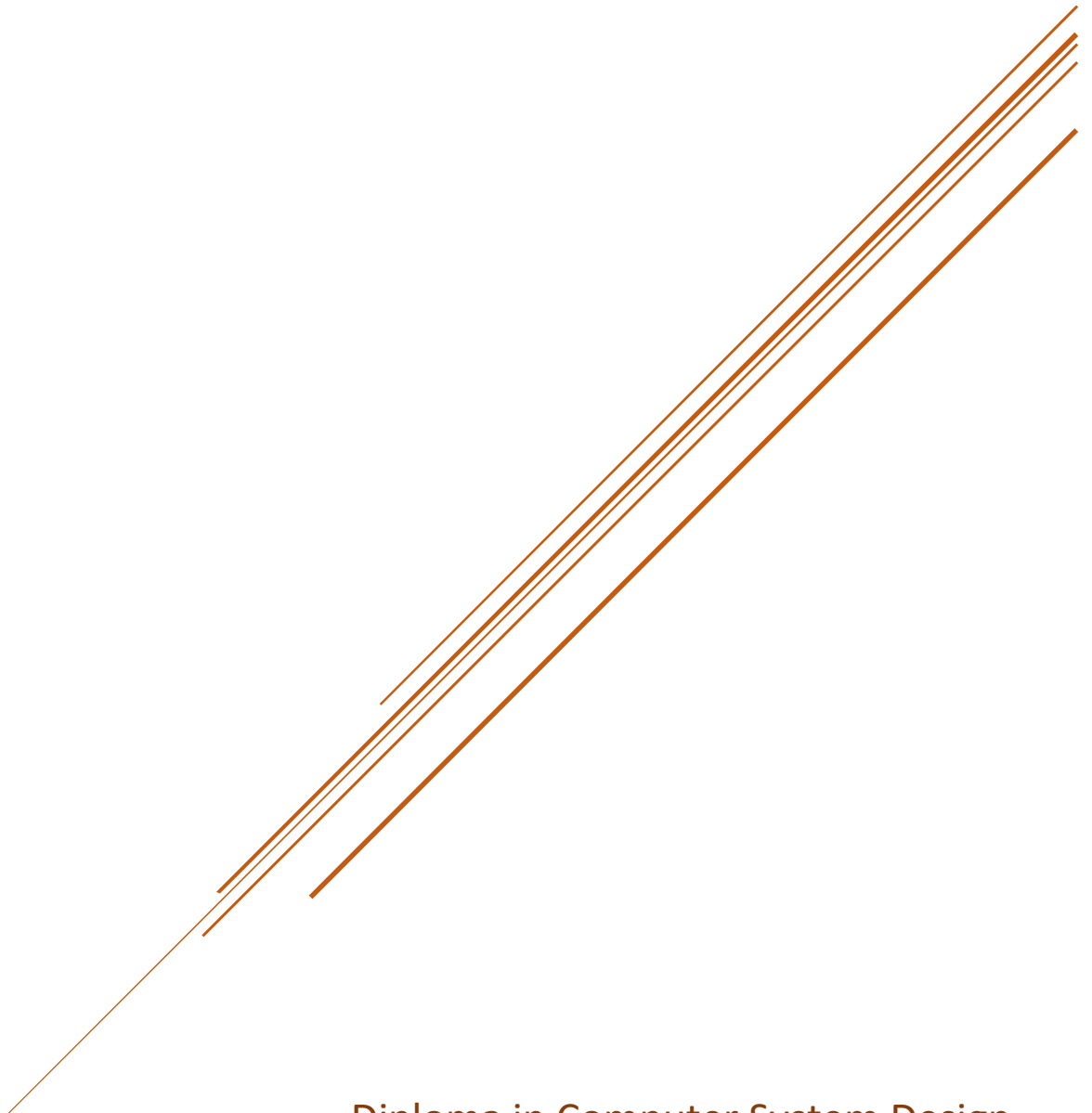


DINEWAY - RESTAURENT MANAGEMENT SYSTEM

Project Report



Diploma in Computer System Design
National Institute of Business Management – Colombo 07

Declaration

We hereby declare that the work presented in this project report, titled “**Dineway Restaurant Management System**”, is our original effort carried out under the guidance and supervision of **Mrs. Chandula Rajapaksa** as part of the Diploma in Computer System Design, Batch **DCSD23.1F**, at the National Institute of Business Management (NIBM).

We confirm that:

1. This project is the result of our own independent work and has not been submitted elsewhere for the award of any other diploma, degree, or qualification.
2. All sources of information, data, and materials used in the preparation of this report have been duly acknowledged and referenced.
3. We have adhered to the ethical standards and academic guidelines prescribed for the completion of this project.

The members of the project team are as follows:

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We take full responsibility for the content and outcomes of this project report.

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Chapter 1: Introduction

1.1 Overview

The food and hospitality industry in Sri Lanka is experiencing rapid growth, with increasing demand for efficiency, convenience, and customer-centric services. Restaurants must balance the delivery of quality food with streamlined operations to remain competitive.

Dineway Restaurant, located at Walpola Junction, Angoda, has operated successfully for over 12 years, offering Sri Lankan, Indian, and Chinese cuisines. With multiple branches across the country, Dineway has established itself as a recognized brand in the local dining market.

The restaurant operates daily from **11:00 a.m. to 10:00 p.m.**, catering to both dine-in and takeaway customers. Its **open kitchen concept** allows customers to observe food preparation, enhancing transparency and trust. A cozy seating arrangement, complemented by soft music, provides a welcoming dining atmosphere.

Beyond traditional services, Dineway offers **event catering** and has expanded its reach through integration with **Uber Eats**, making its food accessible to customers beyond physical outlets. The restaurant employs approximately **26 staff members**, including chefs, waiters, cashiers, delivery drivers, and support staff, managed by the Restaurant Manager.

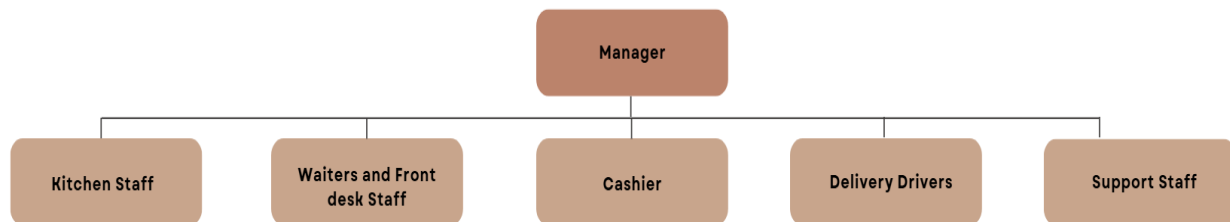
However, despite partial computerization in areas such as billing and ordering, the restaurant still faces operational inefficiencies. Systems work in isolation, and certain processes are manual, creating delays, errors, and communication gaps.

To address these challenges, our project team conducted a **Business Analysis study** of Dineway's current operations, identified existing drawbacks, and proposed an updated **Restaurant Management System (RMS)**. The system design focuses on automation, integration, and improved communication, with the aim of delivering a more efficient and customer-friendly operational model.



1.2 Organization Structure

The organizational hierarchy of Dineway is designed to ensure smooth coordination between different operational units. At the top is the **Restaurant Manager**, who acts as the decision-maker and ensures that customer satisfaction and profitability goals are met.



The main components of the organizational structure include:

- **Manager** - Provides leadership, supervises staff, oversees finances, and ensures quality and compliance with health and safety regulations.
- **Chefs and Kitchen Staff** - Responsible for preparing meals, managing kitchen supplies, and maintaining food quality standards.
- **Waiters and Front Desk Staff** - Handle customer orders, reservations, and ensure proper service delivery.
- **Delivery Drivers** - Coordinate takeaway and online orders, particularly those made through Uber Eats and the proposed online ordering system.
- **Cashiers** - Manage billing, payment processing, and maintain financial accuracy.
- **Support Staff** - Assist in cleaning, inventory handling, and other auxiliary tasks essential for smooth operations.

This structure provides clear accountability but lacks **technological integration** across departments. For example, *while the billing staff rely on a computerized system, the kitchen and delivery staff often depend on manual updates or verbal communication. This disconnect creates operational delays and inconsistencies.*

1.3 Current Operations in the Organization

Currently, the restaurant uses a **partially automated system** that supports only a few functions:

- **Computerized Ordering System** - Allows staff to enter dine-in and takeaway orders.
- **Billing System** - Generates customer bills and handles payments.
- **Employee Registration System** - Stores employee data but does not extend to payroll or attendance tracking.
- **Delivery Management** - Integrated with Uber Eats to handle external delivery requests.
- **Sales Reporting** - Generates daily reports based on billing records.

Although these systems contribute to the efficiency of certain processes, they operate **in isolation** rather than as a fully integrated solution. As a result:

- Information flow between the kitchen and cashier is inefficient.
- Customers frequently face delays in receiving updates about their takeaway orders.
- Payroll is still handled manually, creating room for errors.
- Inventory and supplier management are not computerized, making it harder to track stock levels.

This semi-automated environment is no longer sufficient for a growing restaurant that aims to provide fast, reliable, and consistent service.

1.4 Problem Definition

Despite the partial use of computerized systems, several **operational challenges** persist at Dineway:

1. **Manual Payroll Management**
Salaries are calculated and recorded manually. This process is time-consuming, prone to human error, and does not account for deductions, bonuses, or overtime efficiently.
2. **Revenue Reporting Difficulties**
Although daily sales are tracked, there is no consolidated reporting mechanism to analyze performance trends, profitability, or menu item popularity. This lack of insights limits management's ability to make informed business decisions.
3. **Communication Gap Between Front Staff and Kitchen**
The ordering process is not seamlessly connected with the kitchen. Miscommunication can result in order delays, incorrect dishes being prepared, or poor coordination during peak hours.
4. **Takeaway Order Inefficiency**
Customers who place takeaway orders often face delays and have to repeatedly ask about the status of their food. This not only creates frustration but also reduces the likelihood of repeat customers.

5. Lack of Integrated Inventory Management

The absence of an automated stock tracking system leads to inefficiencies in monitoring raw materials, supplier deliveries, and reordering, which can result in wastage or shortages.

These challenges clearly indicate the **need for an integrated Restaurant Management System** that combines payroll, order management, inventory control, and customer communication into a single streamlined platform.

Chapter 2: Methodology

A systematic and structured methodology is critical to the success of any project, particularly in the context of developing a comprehensive **Restaurant Management System (RMS)**. The methodology not only defines the process for gathering requirements but also ensures that the design and implementation stages are grounded in real operational challenges faced by the organization.

This chapter outlines the **data collection methods** employed by us, the rationale behind their selection, and the steps taken to ensure accuracy, relevance, and reliability of the gathered information.

2.1 Data Collection Methods and Justification

In order to gain a thorough understanding of the existing operations at **Dineway Restaurant**, we utilized a combination of **qualitative** and **quantitative** data collection techniques. The objective was to capture the strengths and weaknesses of the current system while identifying areas where improvements could be made through technological intervention.

The following methods were used:

1. Site Visits and Direct Observation

We conducted multiple visits to the restaurant premises to observe the daily workflow. This allowed us to:

- Monitor how orders were received and processed.
- Identify communication flows between the cashier, kitchen, and delivery staff.
- Understand the time taken for each process, including billing, serving, and delivery.
- Note how inventory and stock handling were managed.

Justification: Direct observation provided a real-time and unbiased perspective of the operational environment. It highlighted inefficiencies, such as delays in kitchen communication and manual handling of payroll.

2. Employee and Manager Interviews

Structured and semi-structured interviews were conducted with key stakeholders, including the restaurant manager, chefs, waiters, cashiers, and delivery staff. The questions focused on:

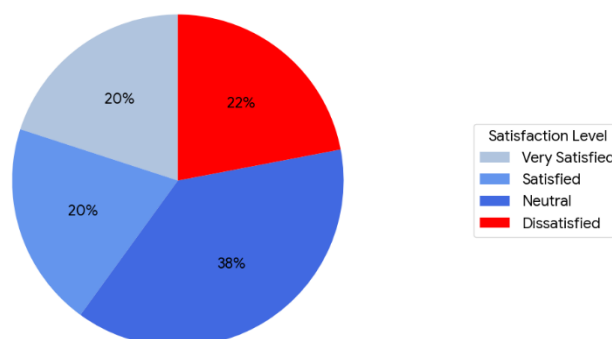
- Challenges they face in their respective roles.
- The level of satisfaction with the existing computerized systems.
- Suggestions for improvements or new features that could simplify their work.

Justification: Interviews gave us valuable **qualitative insights** into employee frustrations and expectations. *For instance, waiters emphasized the need for faster communication with the kitchen, while manager highlighted difficulties in generating consolidated financial reports.*

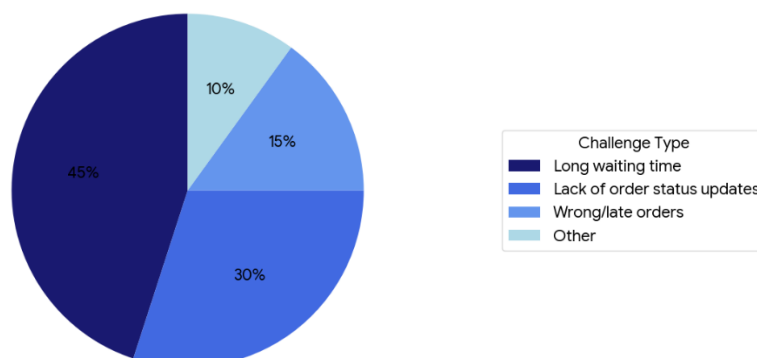
3. Customer Surveys

Questionnaires were distributed to a sample of dine-in and takeaway customers. Questions focused on:

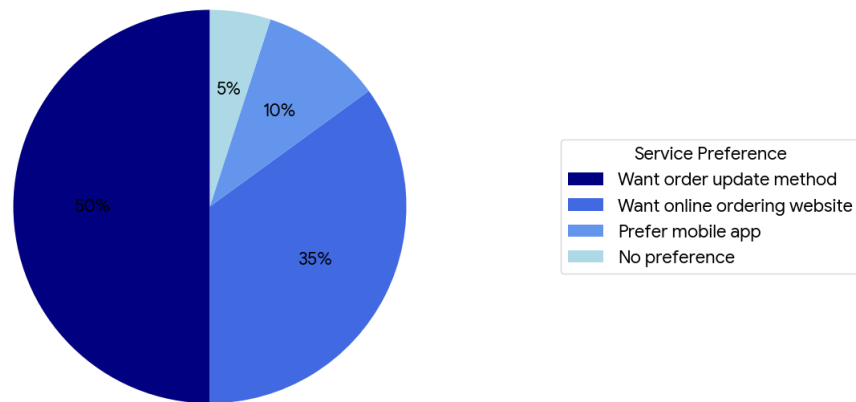
- Overall satisfaction with service speed and accuracy.



- Issues faced while placing takeaway orders.



- Expectations for additional services, such as order tracking or online ordering platforms.



Justification: Customers are the ultimate users of the system's output. Their feedback provided direct insight into service gaps - *especially the frustration caused by waiting for takeaway orders without updates.*

4. Document and System Review

Existing documents such as billing records, payroll logs, and sales reports were examined. In addition, the current computerized billing and ordering systems were reviewed for scope and functionality.

Justification: Document analysis provided **quantitative data** regarding sales, staffing, and financial management. It also helped *confirm inconsistencies, such as errors in manual payroll calculations.*

2.2 Approach to System Design

Based on the information collected, we applied a **Business Analysis oriented design approach**. This included:

1. **Requirement Gathering** - Identifying functional and non-functional requirements through observation, interviews, and surveys.
2. **Gap Analysis** - Comparing the current system (as-is) with the desired system (to-be) to identify areas for improvement.
3. **System Modeling** - Designing diagrams such as **use case diagrams, class diagrams, sequence diagrams, and ERDs** to represent proposed solutions.

4. **Process Documentation** - Preparing workflows, file structures, and report layouts to clearly document how the proposed system would operate.
5. **Validation with Stakeholders** - Reviewing proposed features with the restaurant manager and staff to ensure alignment with actual needs.

2.3 Justification of Methodology

By combining **observations, interviews, surveys, and document reviews**, we ensured that the requirements gathered reflected both operational needs and customer expectations. Using a **requirements-driven, design-focused approach** allowed us to:

- Capture all critical system gaps.
- Propose solutions that are realistic and user-focused.
- Document a comprehensive system design that could guide future developers during implementation.

Chapter 3: System Study and Proposed Solution

The purpose of this chapter is to provide a comprehensive analysis of the existing system at Dineway Restaurant, identify its shortcomings, and introduce the proposed **Restaurant Management System (RMS)**. The chapter also outlines the potential risks involved in the project and highlights the key benefits expected from the new system.

3.1 Drawbacks of the Current System

Our analysis revealed several critical inefficiencies in the existing system:

1. Manual Payroll Entries

- Employee salaries are calculated manually each month.
- This process is time-consuming and prone to human error.
- No link exists between employee attendance and salary calculation, making it difficult to track overtime, leave, or deductions accurately.

2. Revenue Reporting Limitations

- The current system only provides basic daily sales reports.
- Advanced insights (e.g., sales trends, most popular items, long-term revenue patterns) are unavailable.
- Management lacks the data required for strategic decisions.

3. Communication Gap Between Cashier/Front Staff and Kitchen

- Orders are relayed verbally or using printed slips carried by waiters.
- This leads to delays, miscommunication, and occasional preparation of incorrect dishes.

4. Inefficient Takeaway Order Process

- Customers who place takeaway orders must frequently ask staff about order readiness.
- This creates customer dissatisfaction and additional workload for front staff.

5. Lack of Inventory Management

- Stock usage and supplier deliveries are tracked manually.
- There is no systematic way to monitor available ingredients, leading to shortages or wastage.

These drawbacks illustrate the fragmented nature of the current system and highlight the need for an **integrated solution** that connects all critical areas of restaurant operations.

3.2 Introducing the New System

To address the issues identified, our team **proposed a new Restaurant Management System (RMS)**. The system is designed as an integrated solution covering payroll, reporting, order management, communication, inventory, and online ordering.

Proposed Solutions Aligned to Current Drawbacks:

1. Automated Payroll and Attendance Tracking

- A payroll module is proposed within the **Manager's Dashboard**.
- Attendance recording functionality is included to log work hours, late arrivals, and absences.
- Salaries are automatically calculated based on attendance data, reducing errors and saving time.

2. **Automated Revenue Reporting**

- The Manager's Dashboard includes a feature to generate revenue reports automatically.
- Reports can be produced daily, weekly, or monthly, allowing management to track performance and trends.

3. **Improved Kitchen Communication**

- We proposed linking the cashier's receipt printer to a **secondary printer in the kitchen**.
- Once an order is placed and paid for, order details are automatically printed in the kitchen, ensuring immediate communication without manual effort.

4. **Takeaway Order Tracking via LED Display**

- An **LED order status board** is proposed for the waiting area.
- Orders are displayed by number, with statuses such as "Preparing" and "Ready for Pickup."
- This reduces customer frustration and minimizes repeated inquiries.

5. **Inventory Management System**

- A shared inventory module is proposed for both the Manager and Kitchen Staff.
- Inventory usage can be updated daily, with the system tracking stock levels in real time.
- Alerts are generated for low stock, ensuring timely reordering and reducing wastage.

6. **Online Ordering Website**

- We proposed the development of a dedicated online ordering platform.
- Customers can place orders directly, bypassing third-party apps like Uber Eats.
- This improves customer convenience while reducing third-party commission costs for the restaurant.

Through these solutions, the proposed RMS directly addresses every drawback of the current system, ensuring a smoother, faster, and more customer-friendly restaurant operation.

3.3 New System Overview

The **proposed RMS** is designed as a **centralized software platform** that integrates all major restaurant functions into a single system.

Core Modules of the Proposed RMS:

1. Manager's Dashboard

- Provides a consolidated view of key metrics such as daily revenue, active orders, and inventory levels.
- Includes automated payroll and attendance tracking.
- Generates detailed revenue and performance reports.

2. Employee Management Module

- Tracks attendance and shift schedules.
- Automatically calculates employee salaries based on attendance records.
- Stores employee profiles for easy reference.

3. Order and Kitchen Communication Module

- Connects the cashier's receipt printer to the kitchen.
- Ensures order details are instantly available to kitchen staff once payment is processed.

4. Customer Order Tracking Display

- LED board in the waiting area displays order numbers and their preparation status.
- Updates are controlled by front staff for real-time accuracy.

5. Inventory Management Module

- Records inventory usage and stock arrivals.
- Generates low-stock alerts and provides supplier management functionality.
- Accessible by both the Manager and Kitchen Staff.

6. Online Ordering Website

- Allows customers to browse menus, place orders, and make payments online.
- Provides digital receipts and order tracking for customer convenience.

System Characteristics:

- **Integration** - All modules are interconnected to avoid duplication of work.
- **User-Friendliness** - Designed with simple interfaces for staff with minimal IT background.
- **Scalability** - Capable of expanding with future restaurant branches.

3.4 Risk Assessment

Implementing a new system comes with certain risks. The following risks and mitigation strategies were identified:

1. Technology Integration Issues

- *Risk:* Compatibility issues may arise when integrating the new system with existing hardware or third-party services.
- *Mitigation:* Conducting extensive system testing before deployment and ensuring proper configuration.

2. Project Timeline Delays

- *Risk:* Delays in software development or unforeseen external factors could push back the project timeline.
- *Mitigation:* Adopting an **iterative development model** and maintaining regular progress reviews to stay on track.

3. Staff Resistance to Change

- *Risk:* Employees may resist adopting the new system due to lack of familiarity or fear of complexity.
- *Mitigation:* Conducting training sessions, providing user manuals, and ensuring user-friendly interfaces.

4. Data Security Concerns

- *Risk:* Storing sensitive data such as employee records and customer details may pose security risks.
- *Mitigation:* Implementing encryption, access control, and regular backups to secure the system.

5. Cost Overruns

- *Risk:* Unplanned expenses during development or implementation may exceed the budget.
- *Mitigation:* Detailed budgeting, phased implementation, and cost monitoring throughout the project.

3.5 Key Benefits of the Proposed System

The proposed RMS offers several benefits to both management and customers:

- **Efficiency Boost** - Automation of payroll, order processing, and inventory tracking reduces manual workload and errors.
- **Data-Driven Decision Making** - Real-time reports and analytics support strategic planning, menu optimization, and cost control.
- **Improved Staff Productivity** - Biometric attendance and shift scheduling streamline workforce management.
- **Enhanced Customer Experience** - Faster service, takeaway order tracking, and online ordering improve satisfaction and loyalty.
- **Reduced Costs** - Optimized inventory reduces wastage, while direct online ordering minimizes third-party commissions.
- **Scalability** - The RMS can easily adapt to new branches or future restaurant expansion.

In summary, the new system is designed not only to solve current challenges but also to **future-proof Dineway's operations** in an increasingly competitive restaurant industry.

Chapter 4: System Analysis and Design

The system analysis and design phase focuses on transforming business requirements into a structured system architecture. For the proposed Restaurant Management System (RMS), a full set of **UML diagrams and database models** were developed to represent system functionality, workflows, and data relationships.

Since this report serves as documentation of the study rather than a technical design manual, only a **high-level overview** of the system architecture is presented here. The complete diagrams - including sequence diagrams, class diagrams, entity-relationship diagrams, and detailed user interface mockups - were created using **UML diagramming tools** during the project but are not attached in full to this document.

4.1 Proposed Use Case Diagram

The **Use Case Diagram** illustrates how different users (actors) interact with the RMS. The system identifies three primary actors:

- **Manager** - Oversees payroll, reports, inventory, and employee management.
- **Employees** (Waiters, Chefs, Delivery Staff, Cashiers) - Handle orders, update status, and interact with customers.
- **Customers** - Place dine-in, takeaway, or online orders, and provide feedback.

Key Use Cases include:

- Place Order (dine-in, takeaway, or online)
- Manage Inventory (add, update, reorder items)
- Generate Reports (sales, payroll, customer feedback)
- Process Payments (cash, card, online)
- Manage Employee Payroll
- Track Order Status (for customers and staff)
- Manage Reservations (table bookings)

Explanation:

This diagram ensures clarity in defining *what the system should do* rather than *how it will be implemented*. It highlights all possible interactions and ensures that system requirements cover all critical functions.

4.2 Class Diagram

The **Class Diagram** represents the object-oriented structure of the system. Key classes include:

- **Manager** - Attributes: NIC, Name, Address, Contact, Email; Methods: approvePayroll(), generateReport().
- **Employee** - Attributes: EmployeeID, Name, Position, Salary; Methods: markAttendance(), updateSchedule().
- **Customer** - Attributes: CustomerID, Name, Contact, Address; Methods: placeOrder(), giveFeedback().
- **Order** - Attributes: OrderID, Date, Time, Amount, Status; Methods: updateStatus(), calculateTotal().
- **Menu** - Attributes: MenuID, ItemName, Description, Price; Methods: updateItem(), deleteItem().
- **Inventory** - Attributes: InventoryID, ItemName, Quantity, Supplier; Methods: updateStock(), reorderItem().
- **Payment** - Attributes: PaymentID, Amount, Date, Type; Methods: processPayment().

Explanation:

By defining these classes, the system can be implemented using object-oriented programming principles. This ensures modularity, code reusability, and maintainability.

4.3 Sequence Diagrams

Sequence diagrams describe the **flow of events** for specific use cases:

1. Order Placement Sequence

- Customer → Waiter/Cashier → System → Kitchen → Customer
- The order is entered, recorded in the system, sent to the kitchen, prepared, and then updated for serving/delivery.

2. Payroll Sequence

- Manager → System → Employee Database → Payroll Module → Manager
- Salaries are calculated automatically, deductions applied, and reports generated for manager approval.

3. Online Order Sequence

- Customer → Website → System → Delivery Driver
- The system validates order details, updates inventory, and notifies delivery staff.

A comprehensive set of sequence diagrams was developed to illustrate how the system handles various operational scenarios. Separate diagrams were created for **each instance of system use**, including dine-in order placement, takeaway processing, payroll calculation, revenue reporting, and online order management.

These diagrams demonstrate the logical flow of information and communication between different modules of the RMS. Although the full set of sequence diagrams is not included in this report, they collectively validate that the proposed system design eliminates the inefficiencies observed in the current system.

Explanation:

These diagrams help visualize the communication between system components and ensure that no step is overlooked in system design.

4.4 Entity Relationship Diagram (ERD)

The ERD illustrates the **data relationships** in the RMS. Key entities include:

- **Employee (E_ID, NIC, Name, Position, Salary, Contact)**
- **Customer (C_ID, Name, NIC, Contact, Address)**
- **Order (O_ID, Item, Date, Amount, CustomerID, EmployeeID)**
- **Menu (M_ID, ItemName, Description, Price)**
- **Inventory (InventoryID, ItemName, Quantity, SupplierID)**
- **Supplier (SupplierID, Name, Contact, Address)**
- **Payment (P_ID, Amount, Date, OrderID)**
- **Reservation (R_ID, Date, Time, TableNo, CustomerID)**
- **Feedback (F_ID, Comment, Rating, CustomerID)**

Relationships:

- One **Customer** can place multiple **Orders**.
- One **Order** is linked to one **Payment**.
- One **Employee** can handle multiple **Orders**.
- One **Supplier** provides multiple **Inventory Items**.

The ERD was designed to provide a logical representation of the database structure. It includes multiple entities and **extensive relationships** among them, such as customer-to-order, employee-to-attendance, inventory to supplier, and order to payment.

While only a simplified overview is referenced here, the full ERD developed during the project captures all required data relationships to support integrated restaurant operations.

Explanation:

The ERD provides a solid foundation for database design by clarifying data relationships, ensuring normalization, and eliminating redundancy.

4.5 File Handling

As part of the system design, the required file and database structures were modeled to ensure that all operational data is stored and managed efficiently. These structures were documented using **UML and ERD tools**, covering all critical data flows within the restaurant.

The proposed system design specifies data handling for:

1. **Manager Table** - Stores manager details.
2. **Employee Table** - Stores employee details, positions, and NIC.
3. **Menu Table** - Contains all menu items and their descriptions.
4. **Order Table** - Tracks order details, time, and assigned staff.
5. **Inventory Table** - Manages items, quantities, and supplier details.
6. **Supplier Table** - Maintains supplier details and their associated items.
7. **Customer Table** - Holds customer records, including reservations and feedback.
8. **Payment Table** - Records all transactions.
9. **Salaries Table** - Stores payroll records, bonuses, and loans.
10. **Reservation Table** - Manages table bookings and availability.
11. **Feedback Table** - Collects customer feedback and ratings.

In addition, the database design establishes proper **primary keys, foreign keys, and relationships** across these tables, ensuring referential integrity and smooth integration of all modules within the system.

Explanation:

By defining these file structures at the design stage, the proposed system ensures that data will be consistently organized, accessible for reporting, and scalable for future extensions.

4.6 User Interfaces

Interactive user interfaces were prototyped for all modules of the system using **Figma UI Designing Tool**. The RMS includes a set of user-friendly interfaces:

- **Login Screen** - Secure login for Manager, Employees, and Admin.
- **Dashboard** - Provides real-time metrics (active orders, revenue, pending deliveries).
- **Order Management Interface** - Allows waiters/cashiers to enter and update orders.
- **Kitchen Display System** - Displays incoming orders for chefs in real-time.
- **Inventory Management Screen** - For stock managers to update and reorder supplies.
- **Payroll and Employee Management Screen** - For managing salaries, attendance, and schedules.
- **Customer Feedback Form** - Digital form for customers to leave ratings and comments.
- **Online Ordering Website UI** - Customer-facing web application for placing orders, tracking deliveries, and making payments.

***Special attention was given to usability, with features such as popup notifications, error messages, and confirmation alerts included to improve user experience.**

Explanation:

Special emphasis was placed on making the system intuitive, reducing the need for extensive training, and ensuring that even non-technical staff can operate the system efficiently.

4.7 Report Layouts

Reports are an essential part of the RMS for both management and staff decision-making. The following report layouts were designed:

1. **Monthly Salary Report** - Lists employee salaries, deductions, and bonuses.
2. **Daily Sales Report** - Summarizes transactions for a given day.
3. **Inventory Status Report** - Shows stock levels, low-stock alerts, and reorder history.
4. **Customer Feedback Report** - Summarizes ratings, comments, and service improvement areas.
5. **Delivery Report** - Shows completed and pending deliveries with driver assignments.

Explanation:

Each report was formatted to provide clear and actionable insights. Reports can be generated in both printable and digital formats for convenience.

4.8 Summary of System Design

In summary, the system design phase produced a complete architecture for the proposed RMS, documented through UML diagrams, ERD models, user interface mockups, and report templates. Although this chapter only provides a brief overview, the detailed diagrams created using **UML diagramming tools** are available as part of the project deliverables for future reference.

Chapter 5: Conclusion

5.1 Summary

This project analyzed the current operations of **Dineway Restaurant** and proposed a comprehensive **Restaurant Management System (RMS)** to address the inefficiencies observed. Through site visits, interviews, surveys, and document reviews, the study highlighted several critical challenges, including manual payroll processes, limited reporting, weak communication between cashier and kitchen staff, lack of inventory management, and limited order status transparency for customers.

To resolve these challenges, a **proposed system design** was developed with integrated modules for payroll and attendance, automated revenue reporting, order communication via linked receipt printers, takeaway order tracking through LED displays, inventory management, and an online ordering platform.

The system architecture was represented through **UML-based diagrams** (use case diagrams, sequence diagrams, class diagrams, and ERDs) and interactive user interface designs. Report layouts were also designed to provide meaningful insights to management. While this chapter only outlines the high-level outcomes, the detailed designs are available in the complete system documentation.

5.2 Final Remarks

The proposed RMS demonstrates how a structured approach to system study and design can provide effective solutions for real-world operational challenges. By adopting such a system, Dineway Restaurant can expect improved efficiency, reduced errors, enhanced customer satisfaction, and stronger support for long-term growth.